Ajay Venkatesh

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Skills

- **Programming Languages:** Python, Java, Scala, R, SQL, JavaScript (ES6)
- **Big Data & Processing:** Pandas, NumPy, PySpark, Apache Spark, Snowflake, Databricks, Kafka
- **Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, Dataverse
- **Cloud & Warehouses:** AWS (Redshift, S3, EC2, Lambda, Glue, CloudWatch, SageMaker, CDK), GCP (BigQuery, Dataflow), Azure (Data Lake, Data Factory, Function Apps, Storage Blobs)
- **Tools & BI:** Git, Docker, Kubernetes, Jupyter, Postman, Power BI, AWS QuickSight, Visual Studio, Cursor

Experience

- **CampusMesh** *Jan 2026 – Present*
  Software Engineer New York, US
  - Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
  - Deployed **serverless architecture** on AWS (**Lambda, API Gateway, S3, CloudFront**) across **4 environments** with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
  - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- **Amazon** *May 2025 – Aug 2025*
  Software Engineer Intern Seattle, US
  - Architected a secure data console for Amazon fulfillment center teams to query image/video evidence across FCs; presented **system design** for approval and set up Role-Based Access **(4+ roles)** via **OAuth, Midway SSO, HELIS**.
  - Handled large-scale **image and video evidence data** from fulfillment centers, storing assets in **AWS S3** with optimized access patterns and retrieving via **pre-signed URLs** for low-latency display in the frontend.
  - Engineered distributed backend services in **Java** (Coral, Smithy, Dagger, CBOR); applied **low-level design patterns**, cutting execution time by **35%** with unit/integration testing via **Mockito**.
- **New York University (Novel AI Technologies, Inc.)** *Aug 2024 – Present*
  Data Engineer New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android**, ingesting real-time wearable telemetry into cloud storage for downstream AI model inference on health metrics and food images.
  - Engineered **ETL/ELT workflows** with **Python, PySpark, SQL**, transforming **NoSQL** event data into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records.
  - Built **AWS QuickSight** dashboards with **role-based access** to analyze collected user health data and surface **risk and health scores**; embedded the dashboards into the customer-facing website, generating **$50K+** revenue across **6 clients**.
- **PricewaterhouseCoopers (PwC)** *Aug 2021 – Aug 2024*
  Senior Associate Bengaluru
  - Developed a customer **onboarding application** on **Microsoft CRM**, **Microsoft Azure**, and **Dataverse** with an **event-driven architecture** powering downstream business workflows.
  - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction from unstructured documents, storing results in a relational **database** and **Azure Storage Blobs**, reducing processing time by **75%**.
  - Optimized batch ingestion with **parallel processing** on **100K-row** Excel feeds daily; reduced execution from **5 hours to 1 hour**.
  - **Partnered with clients** to translate business requirements into technical deliverables under **Agile / Scrum** workflows; **mentored junior engineers** and led structured **code reviews**, raising team delivery quality and reducing production defects by **25%**.
  - Built **Power BI** dashboards joining datasets across multiple relational databases, improving reporting efficiency by **40%**.

Projects

- **Flight Delay Prediction (Big Data)** | *Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Matplotlib, Plotly*
  - Built a distributed big data pipeline over **Hadoop HDFS** ingesting **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
  - Trained **Spark MLlib** classification models to predict delay severity and surfaced feature importance and delay distributions via **Seaborn, Matplotlib, and Plotly** visualizations.
- **InviLite (Cloud-Native Data Pipeline)** | *AWS Kinesis, Lambda, S3, Glue, DynamoDB, SageMaker, Bedrock, FAISS, Python*
  - Architected a cloud-native data pipeline (**Kinesis, Lambda, S3, Glue, DynamoDB**) ingesting structured and unstructured enterprise data into governed ML pipelines with schema validation and access controls.
  - Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.

Education

New York University, New York, NY *Aug 2024 – May 2026*
Master of Science in Computer Engineering (GPA: 3.9/4.0)
Coursework: Big Data, System Design, Machine Learning, Deep Learning, Data Structures, Advanced Java

Visvesvaraya Technological University, India *Jun 2017 – Aug 2021*
Bachelor of Engineering in Electronics and Communication (GPA: 8.9/10)

Certifications

AWS Cloud Practitioner | **Azure Data Fundamentals** | **Azure Data Scientist Associate**
Power BI Data Analyst Associate | Stanford Machine Learning (Coursera)